

Air Quality Index Prediction Using Machine Learning

Project Overview

This project focuses on predicting the Air Quality Index (AQI) using various environmental and pollution parameters. The objective is to analyze air pollution data and build a machine learning model capable of estimating AQI levels based on pollutant concentrations and environmental factors.

Objective

- Analyze air quality data.
- Perform data preprocessing and visualization.
- Train a machine learning model for AQI prediction.
- Evaluate model performance using standard metrics.
- Understand the impact of pollutants on air quality.

Dataset Description

The dataset contains the following attributes:

Column Name	Description
Place	Location name
PM2.5	Fine particulate matter concentration
PM10	Coarse particulate matter concentration
NO2	Nitrogen Dioxide concentration
SO2	Sulfur Dioxide concentration
CO	Carbon Monoxide concentration
O3	Ozone concentration
Temperature	Ambient temperature
Humidity	Relative humidity
Traffic_Level	Traffic intensity
Industrial_Area	Presence of industrial activity
Population_Density	Population density
AQI	Air Quality Index (Target Variable)

Technologies Used

- Python
- Pandas

- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook

Project Workflow

1. Data Collection

Collected air quality and environmental data for multiple locations.

2. Data Preprocessing

- Handling missing values
- Converting date formats
- Encoding categorical variables
- Feature selection

3. Exploratory Data Analysis

Various visualizations were created to understand data patterns and pollutant relationships.

4. Model Training

A Linear Regression model was trained using pollution and environmental features.

5. Prediction

The trained model predicts AQI values based on input parameters.

6. Evaluation

Model performance was evaluated using:

- Mean Absolute Error (MAE)
- R² Score

Visualizations

The following graphs were generated:

- Average AQI by Place
- Actual vs Predicted AQI

Machine Learning Model

Algorithm Used

Linear Regression

Features Used

- PM2.5
- PM10
- NO2
- SO2
- CO
- O3
- Temperature
- Humidity
- Traffic Level
- Industrial Area
- Population Density

Results

The model successfully learned relationships between pollutant concentrations and AQI values.

Key observations:

- Higher PM2.5 and PM10 levels significantly increase AQI.
- Traffic and industrial activities contribute to poor air quality.
- Environmental factors influence AQI variations across locations.
- The model achieved satisfactory prediction performance.

Conclusion

This project demonstrates how machine learning can be used to predict Air Quality Index values using environmental and pollution parameters. The analysis helps understand air pollution trends and supports better environmental decision-making.